1 Introduction

Lecture 10 is about information complexity, but we will start with communication complexity.

2 Communication Complexity

We start with a function

$$f : X \times Y \to Z$$

2.1 Examples

We have $X, Y = \{0,1\}^n$, $Z = \{0,1\}$, and

$$f : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}$$

Player 1 gets $x \in X$ and player 2 gets $y \in Y$. They have to compute $f(x,y)$

Initially, the plays don't know each other's input. They run a communication complexity protocol. At the end, they both have to know the answer $f(x,y)$.

Each play has unlimited computational power. The issue is communication: How many communication bits the plays have to exchange between them in order to determine the answer.

Three examples of problems:

1. **Equality:** Is $x = y$?
2. **Set Disjointness:** Is there a coordinate i, such that, $x_i = y_i = 1$?
3. **Inner Product:** Compute $\sum_i x_i \cdot y_i \pmod 2$

Sometimes we consider other computational tasks, with possibly more than one correct answer, rather than computation a function (e.g., search problems, promise problems).

2.2 Deterministic Communication Complexity

Deterministic communication protocols: In each step, one of the players sends one bit to the other player. This bit may depend on the previous communication and on the input of the player who sends it. At the end, they both have to know a value $\pi(x,y)$ (the output of the protocol). We require,

$$\forall x \in X, y \in Y, \quad \pi(x,y) = f(x,y)$$

The communication complexity of a protocol, $\mathrm{CC}(\pi)$, is the maximum (over all inputs) of the number of bits transmitted. The deterministic communication complexity of f, $\mathrm{CC}(f)$, is the minimum communication complexity of a protocol computes f.

2.3 Probabilistic Communication Complexity

The players can use random bits. The protocol is probabilistic. Each bit sent may depend on the random bits. The output $\pi(x, y)$ is now a random variable. We require,

$$\forall x \in X, y \in Y, \quad \mathrm{P}(\pi(x, y) = f(x, y)) \geq 1 - \epsilon$$

ϵ is the probability of error (typically, some small constant).

The communication complexity of a protocol, $\mathrm{CC}(\pi)$, is the **maximum** (over all inputs and random strings) of the number of bits transmitted. The probabilistic communication complexity of f, $\mathrm{CC}(f)$, is the **minimum** communication complexity of a probabilistic protocol that computes f with probability of error ϵ.

Reducing $\epsilon \to \epsilon'$ is easy by repeating the protocol and taking majority.

2.4 Public vs Private Randomness

Public random bits can be viewed by both players. Private random bits can be viewed by only one of the players. In general, we can replace all private randomness with public randomness and the protocol still has the same success probability (trivial).

The other direction is also true: we can replace all public randomness with private randomness, if we allow additional $\mathcal{O}(\log n)$ bits of communication and a small increase in the probability of error (not hard but we will not prove).

We allow both public and private randomness as this will be needed when we talk about information complexity.

2.5 Distributive Communication Complexity

There is an underlying joint distribution $P_{X,Y}$ of the inputs X, Y for the two players. The inputs are random variables, not necessarily independent. The protocol is deterministic. We require

$$P_{X,Y}\big(\pi(X, Y) = f(X, Y)\big) \geq 1 - \epsilon$$

We can also allow probabilistic protocols but this equivalent to the deterministic case because we can always take the random string with maximal success probability. When fixing the random string, we get a deterministic protocol. (A probabilistic protocol is just a convex combination (distribution over) probabilistic protocols).

When discussing distributive communication complexity, we sometimes consider the zero error case. (Similar to deterministic communication complexity when X, Y are ...)

2.6 Input Matrix

It is sometimes helpful to view the function f as a matrix:

$$A_{x,y} = f(x, y)$$

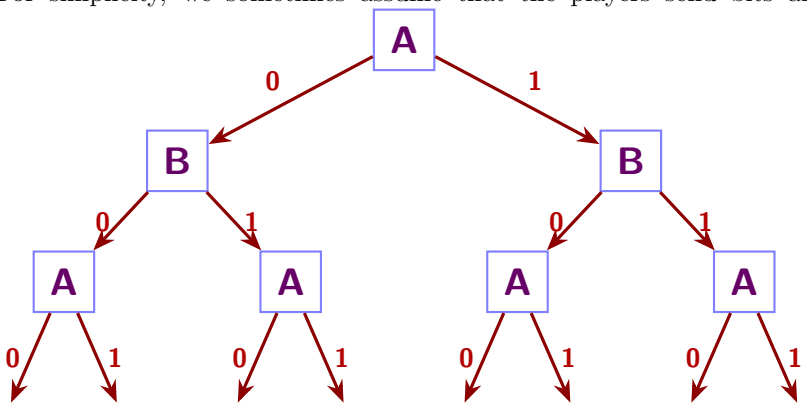
Example: Equality: A is the identity matrix.

2.7 Results

- $\mathrm{CC}(\text{Equality}) = n + 1$
- Upper bound of $n + 1$ true for every function to $\{0, 1\}$
- $\mathrm{CC}_\epsilon(\text{Equality}) = \mathcal{O}(1)$ (public randomness)
- Protocol: Players compare $\sum_i x_i \cdot z_i \pmod 2$ and $\sum_i y_i \cdot z_i \pmod 2$, for random $z_1 \ldots, z_n$. (Can use error correcting codes to reduce randomness)
- $\mathrm{CC}_\epsilon(\text{Equality}) = \mathcal{O}(\log n)$ (private randomness)
- Protocol: Players compare $x \bmod p$ and $y \bmod p$, for a random prime $p \in \{1, \ldots, n^2\}$. (Can use error correcting codes).
- $\mathrm{CC}_\epsilon(\text{Inner-Product}) = \Omega(n)$
- $\mathrm{CC}_\epsilon(\text{Set-Disjointness}) = \Omega(n)$

2.8 Protocol as a Tree

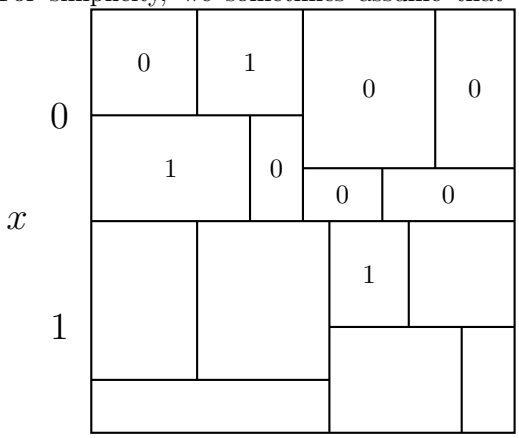
For simplicity, we sometimes assume that the players send bits alternately



Bits sent by Alice may depend on X. Bits sent by Bob may depend on Y.

2.9 Protocol as a Partitioning to Monochromatic Rectangles

For simplicity, we sometimes assume that the players send bits alternately.



3 Information Complexity

3.1 The setup

- X, Y are random inputs to Alice and Bob, with joint distribution $\mu = P_{X,Y}$.
- R: public random string; R_A, R_B: Alice's and Bob's private random strings
- π: probabilistic protocol between Alice and Bob
- $M = (X, Y, R, R_A, R_B)$: transcript of the protocol: all bits transmitted

Let's assume for simplicity that the players send bits in an alternating fashion.

Information Complexity measure the amount of information that the transcript gives about the inputs.

There are two different types:

- External Information Complexity
- Internal Information Complexity

3.2 External Information Complexity

Intuitively, external information complexity is the information that an external observer learns about the inputs X, Y from the transcript M. The formal definition is

$$\begin{aligned}\text{Ext} &= \mathrm{I}\big((X,Y);(M,R)\big)\\ &= \mathrm{I}\big((X,Y);R\big) + \mathrm{I}\big((X,Y);M \mid R\big)\\ &= 0 + \mathrm{I}\big((X,Y);M \mid R\big)\\ &= \mathrm{I}\big((X,Y);M \mid R\big)\end{aligned}$$

because R is independent from X, Y.

3.3 Internal Information Complexity

Intuitively, the information that the players learn about each other's input (X or Y) from the transcript M.

Information that Alice learns about Y from M (conditioned on X):

$$\begin{aligned}\mathrm{I}\big(Y;(M,R,R_A) \mid X\big) &= \mathrm{I}\big(Y;(M,R_A) \mid X,R\big)\\ &= \mathrm{I}\big(Y;M \mid X,R\big) + \mathrm{I}\big(Y;R_A \mid M,X,R\big)\\ &= \mathrm{I}\big(Y;M \mid X,R\big) + 0\end{aligned}$$

The first equality is as before. To show that $\mathrm{I}\big(Y;R_A \mid M,X,R\big) = 0$, we use induction over the transcript: conditioned on M, X, R, we have that Y, R_A are independent.

We define internal information complexity as

$$\text{Int} = \mathrm{I}(Y;M \mid X,R) + \mathrm{I}(X;M \mid Y,R)$$

Intuitively, the sum of the information that Alice learns about Bob and Bob learns about Alice.

3.4 Information Complexity

Now we have two definitions.

$$\begin{aligned}\text{Ext} &= \mathrm{I}\big((X,Y);M \mid R\big)\\ \text{Int} &= \mathrm{I}(Y;M \mid X,R) + \mathrm{I}(X;M \mid Y,R)\end{aligned}$$

3.4.1 Relations

Always,

$$\mathrm{CC}(\pi) \geq \text{Ext} \geq \text{Int}$$

The first inequality is trivial. The second inequality is proved by induction over the transcript.

If X, Y are independent, then

$$\text{Ext} = \text{Int}$$

3.5 Examples

- Alice and Bob send random bits...
- Alice and Bob send constant bits...
- If $X_1 = 0$, Alice sends 0. If $X_1 = 1$, Alice sends 0 with probability $1 - \epsilon$ and 1 with probability ϵ. Both Ext and Int are very small.

4 Zero-Error Information Complexity of AND

4.1 Setup

Let's take $X, Y \in \{0, 1\}$: independent and uniformly distributed. Alice and Bob need to compute $X \wedge Y$ with no error. What is the minimal Ext and Int?

Since X, Y are independent, Ext=Int.

4.1.1 Observation

If Alice and Bob both learn[1] $X \wedge Y$, they each learn the input of the other with probability $\frac{1}{2}$. Together, they learn $\frac{1}{2} + \frac{1}{2} = 1$ bit, just by learning the output. Thus, Int ≥ 1. Hence Ext ≥ 1.

4.1.2 Observation

CC (And) = 2. Hence, Ext=Int≤ 2.

Is there a protocol with Ext, Int ≤ 2?

4.1.3 Protocol

Alice sends X. Bob sends back $X \wedge Y$.

$$\text{Ext} = \text{Int} = 1.5$$

Thus, $1 \leq \text{Ext}, \text{Int} \leq 1.5$

4.2 4 Messages Protocol (ABAB)

The point is that the communication complexity is higher than before, but the information complexity is lower! What the hell?!?

$X, Y \in_R \{0, 1\}$. Compute $X \wedge Y$. We have Ext=Int. What is the minimal Ext, Int?

- If $X = 0$, Alice chooses $i \in_R \{1, 3\}$ (the round in which she reveals $X = 0$, and then the game ends)

[1] So if Alice has 0 and learns the output is 0, she doesn't know what Bob has.

- If $Y = 0$, Bob chooses $j \in_R \{2, 4\}$ (the round in which he reveals $Y = 0$, and then the game ends)

First Round:	$X = 1$	$\rightarrow$	A sends 1	
	$X = 0,\ i = 3$	$\rightarrow$	A sends 1	
	$X = 0,\ i = 1$	$\rightarrow$	A sends 0	Stop
Second Round:	$Y = 1$	$\rightarrow$	B sends 1	
	$Y = 0,\ j = 4$	$\rightarrow$	B sends 1	
	$Y = 0,\ j = 2$	$\rightarrow$	B sends 0	Stop
Third Round:	$X = 1$	$\rightarrow$	A sends 1	
	$X = 0$	$\rightarrow$	A sends 0	Stop
Fourth Round:	$Y = 1$	$\rightarrow$	B sends 1	Stop
	$Y = 0$	$\rightarrow$	B sends 0	Stop

Probability	**Transcript**	**External Observer learned**
$\frac{1}{4}$	0	1
$\frac{3}{4} \cdot \frac{1}{4}$	10	$1 + \left(1 - \mathrm{H}\left(\frac{1}{3}, \frac{2}{3}\right)\right)$
$\frac{3}{4} \cdot \frac{3}{4} \cdot \frac{1}{3}$	110	$1 + \left(1 - \mathrm{H}\left(\frac{1}{3}, \frac{2}{3}\right)\right)$
$\frac{1}{8}$	1110	2
$\frac{1}{4}$	1111	2

$$\text{Ext} = \text{Int} = 1 + \frac{3}{8} \cdot 1 + \frac{3}{8} \cdot \left(1 - \mathrm{H}\left(\frac{1}{3}, \frac{2}{3}\right)\right) < 1.5$$

4.3 Best Protocol

$X, Y \in_R \{0, 1\}$. Compute $X \wedge Y$. Ext = Int. What is the minimal Ext, Int? We take $n \to \infty$. n is even and ABAB...

- If $X = 0$, Alice chooses a random odd $i \in \{1, \ldots, n\}$ (the round in which she reveals $X = 0$, and then the game ends).
- If $Y = 0$, Bob chooses a random even $j \in \{1, \ldots, n\}$ (the round in which he reveals $Y = 0$, and then the game ends).
- If $X = 1$, Alice sends 1 in all rounds. If $X = 0$, Alice sends 1 until round i and 0 in round i and then the protocol stops.
- If $Y = 1$, Bob sends 1 in all rounds. If $Y = 0$, Bob sends 1 until round j and then the protocol stops

The probability for $X = 1$ and $Y = 1$ slowly increases if the protocol didn't stop. The information is slowly released.

We can think about this as two people trying to figure out if they like each other without hurting the other's feelings.

4.4 Continuous Protocol (Equivalent)

$X, Y \in_R \{0,1\}$. Compute $X \wedge Y$. Ext=Int. What is the minimal Ext, Int?

- If $X = 0$, Alice chooses a random $t_A \in [0,1]$ (the time at which she reveals $X = 0$, and then the game ends).
- If $Y = 0$, Bob chooses a random $t_B \in [0,1]$ (the time at which he reveals $Y = 0$, and then the game ends).
- A clock starts to run (continuously) from time 0 to 1.
- If $X = 0$, Alice raises her hand at time t_A. The protocol stops. Answer: 0.
- If $Y = 0$, Bob raises his hand at time t_B. The protocol stops. Answer: 0.
- If no one has raised a hand by time 1, the protocol stops. Answer: 1.
- The probability that $X = 1$ and $Y = 1$ slowly increases if the protocol hasn't stopped. The information is released slowly.

4.5 Reasons to Study Information Complexity

1. Lower bounds for communication complexity and other models
2. Relations to streaming complexity and other models
3. Information theoretic privacy
4. Direct sum problems: Int is additive for two communication tasks
5. Compression problem: If Int, Ext are small, is there a (possibly different) protocol with small CC?